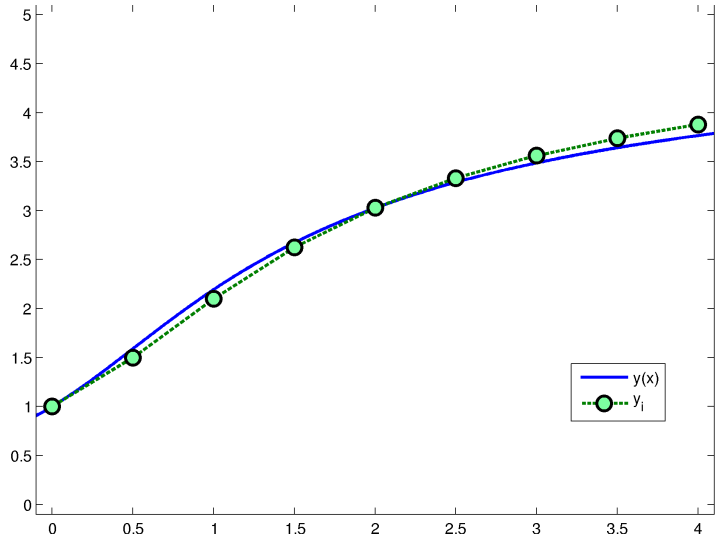
MA385 Part 2: Initial Value Problems

2.3: Error Analysis of Euler's Method

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For more details, see Chapter 6 of Süli and Mayers, *An Introduction to Numerical Analysis*. Some edits were made to these slides after class, mainly changing n to N.

1. General one-step methods

Euler's method, for solving $y'(t) = f(t, y)$ (+ initial condition) is an example of a **one-step method**. These methods have the *general* form:

$$y_{i+1} = y_i + h\Phi(t_i, y_i; h), \quad \text{for } i = 0, 1, \ldots, N-1. \tag{1}$$

To get Euler's method, just take $\Phi(t_i, y_i; h) = f(t_i, y_i)$.

semicolon: h is a parameter (not a variable)

Earlier, we motivated Euler's method with a geometrical argument. An alternative, more mathematical way of deriving Euler's Method is to use a *Truncated Taylor Series*.

The approach will not only motivate Euler's method, but also suggests that at each step the method introduces a (local) error of $h^2 y''(\eta)/2$. (See next slide).

1. General one-step methods

To derive Euler's Method from a Taylor Series, first write down a TS:

$$y(b) = y(a) + (b-a)y'(a) + \tfrac{1}{2}(b-a)^2 y''(\eta) \quad \text{for } \eta \in [a,b]$$

Take $b = t_{i+1}$, $a = t_i$. So $b - a = h$, and

$$y(t_{i+1}) = y(t_i) + h\,y'(t_i) + \tfrac{1}{2}h^2 y''(\eta_i) \quad \eta_i \in [t_i, t_{i+1}]$$

If we neglect the h^2 term, and denote the approximation of $y(t_i)$ as y_i, we get

$$y_{i+1} = y_i + h\,y'(t_i) = y_i + h f(t_i, y_i)$$

where here we have used that $y'(t) = f(t,y)$

2. Two types of error

$y_{i+1} = y_i + h\,\Phi(t_i, y_i; h)$

Definition 2.3.1 (Global Error)

Global Error: $\mathcal{E}_i = y(t_i) - y_i$.

Definition 2.3.2 (Truncation error)

Truncation Error:

$$T_i := \frac{y(t_{i+1}) - y(t_i)}{h} - \Phi(t_i, y(t_i); h). \qquad (2)$$

It can be helpful to think of T_i as representing how much the difference equation differs from the differential equation. For Euler's method, it can be determined using a Taylor Series.

The relationship between the global error and truncation errors is explained in the following (important!) result (also, compare with Picard's Theorem).

Theorem 2.3.1 (Thm 12.1 in Süli & Mayers)

Let $\Phi()$ be *Lipschitz* with constant L. Then

$$|\mathcal{E}_N| \leq T\left(\frac{e^{L(t_N - t_0)} - 1}{L}\right), \tag{3}$$

where $T = \max_{i=0,1,\ldots,N} |T_i|$.

K, a constant.

From the definition of Truncation Error:

$$y(t_{i+1}) = y(t_i) + h\,\Phi(t_i, y(t_i); h) + h\,T_i$$

Method $\quad y_{i+1} = y_i + h\,\Phi(t_i, y_i)$

Subtract $\quad y(t_{i+1}) - y_{i+1} = y(t_i) - y_0 + h\left[\Phi(t_i, y(t_i); h) - \Phi(t, y_i)\right] + h\,T_i$

2. Two types of error

Using $\varepsilon_i = y(t_i) - y_i$

so $\varepsilon_{i+1} = \varepsilon_i + h\left[\Phi(t_i, y(t_i); h) - \Phi(t_i, y_i)\right] + h\tau_i$

Using that Φ is L-Lipschitz, so

$$\left|\Phi(t_i, y(t_i); h) - \Phi(t_i, y_i; h)\right| \le L\left|y(t_i) - y_i\right|$$

$$= L|\varepsilon_i|$$

So $|\varepsilon_{i+1}| \le |\varepsilon_i| + hL\varepsilon_i + h\tau.$

$\Rightarrow |\varepsilon_{i+1}| \le |\varepsilon_i|(1+h) + h\tau.$ | now see exor.

(Finished here Thu, 8 Oct)

3. Analysis of Euler's Method

For Euler's method, we get

1st order.

$$T = \max_{0 \le j \le N} |T_j| \le \frac{h}{2} \max_{t_0 \le t \le t_N} |y''(t)|.$$

Example 2.3.1

Given the problem:

$t_0 = 1$, $y_0 = 1$

$$y' = 1 + t + \frac{y}{t} \quad \text{for } t > 1; \quad y(1) = 1,$$

find an approximation for $y(2)$.

(i) Give an upper bound for the global error taking $N = 4$ (i.e., $h = 1/4$)

(ii) What N should you take to ensure that the global error is no more that 0.1?

3. Analysis of Euler's Method

To answer these questions we need to use (3), which requires that we find L and an upper bound for T. In this instance, L is easy:

(3) $$|\varepsilon_N| \le \frac{T}{L}\left(e^{L(t_N - t_0)} - 1\right).$$

Recall that L is the Lipschitz constant:

$$\Phi(t_i, y_i; h) = f(t_i, y_i)$$

So need L such that

$$|f(t,u) - f(t,v)| \le L|u - v| \quad \text{for all } u, v$$

Here $f(t,y) = 1 + t + \frac{y}{t}$ so

$$|f(t,u) - f(t,v)| \le \left| 1 + t + \frac{u}{t} - 1 - t - \frac{v}{t} \right| \le \frac{1}{t}|u - v|.$$

Since $t \ge 1$ we get $L = 1$.

3. Analysis of Euler's Method

To find T we need an upper bound for $|y''(t)|$ on $[1,2]$, even though we don't know $y(t)$...

But we do know y'(t)! So Differentiate y' to get y''

$$y'(t) = 1 + t + \frac{y}{t} \quad \text{so}$$

$$y''(t) = 0 + 1 + \frac{ty' - y}{t^2} \quad \text{(by quotient rule)}$$

So $y''(t) = 1 + \frac{ty' - y}{t} = 1 + \frac{t(1 + t + \frac{y}{t}) - y}{t^2} = 1 + \frac{1}{t} + 1 + \frac{y - y}{t}$ ($y - y = 0$)

So $y''(t) = 2 + 1/t$. Thus $|y''(t)| \leq 3$.

Note: In other examples the "y" term in the bound may have to be estimated using Picard's Thm from Section 2.1.

3. Analysis of Euler's Method

With these values of L and T, using (3) we find $\mathcal{E}_N \leq 0.644$. In fact, the true answer is 0.43, so we see that (3) is somewhat pessimistic.

. . . . . .

To answer (ii): *What N should you take to ensure that the global error is no more that 0.1?* (We should get $N = 26$. This is not that sharp: $N = 19$ will do).

To answer this, we need $|\varepsilon_N| \leq 0.1$.

So then $\frac{T}{I}\left(e^{L(t_N - t_0)} - 1\right) \leq 0.1$.

That is $\frac{1}{2} h \max_{t_0 \leq t \leq t_N} |y''(t)| \left(e^{L(t_N - t_0)} - 1\right) \leq 0.1$.

We have $L = 1$, $T = 3$, $t_N = 2$, $t_0 = 1$, so solve for h!

Then $N = 1/h$, rounded up to nearest integer.

We are often interested in the *convergence* of a method. That is, is it true that

$$\lim_{h \to 0} y_N = y(t_N)?$$

Or equivalently that,

$$\lim_{h \to 0} \mathcal{E}_N = 0?$$

Given that the global error for Euler's method can be bounded:

$$|\mathcal{E}_N| \leq h \underbrace{\frac{\max |y''(t)|}{2L} \left(e^{L(t_N - t_0)} - 1 \right)}_{K} = hK, \tag{4}$$

we can say it converges.

So now we know, for Euler's method, that $y_N \to y(t_N)$ as $h \to 0$ (or, equivalently, as $N \to \infty$), but how quickly?

Definition 2.3.3

The **order of accuracy** of a numerical method is p if there is a constant K so that

$$|\mathcal{E}_N| \leq Kh^p.$$

So Euler's method is first-order.

The term **order of convergence** is often use instead of **order of accuracy**.

Eg: $p=2$ gives a second-order method.
If $h \sim 10^{-3}$, Euler would give $|\mathcal{E}_N| \leq K10^{-3}$
but a second-order method would give $|\mathcal{E}_N| \leq K10^{-6}$

One of the requirements for convergence is *consistency*:

Definition 2.3.4 (Consistency)

A one-step method $y_{i+1} = y_i + h\Phi(t_i, y_i; h)$, is **consistent** with the differential equation $y'(t) = f\big(t, y(t)\big)$ if

$$f(t, y) \equiv \Phi(t, y; 0).$$

Necc $\lim_{h \to 0} \Phi(t, y; h) = f(t, y)$

This is because

$$\frac{y(t_{i+1}) - y(t_i)}{h} = \Phi(t_i, y_i; h)$$

so $\lim_{h \to 0} \frac{y(t_i + h) - y(t_i)}{h} = y'(t_i) = f(t_i, y_i)$

Next we'll try to develop methods that are of higher order than Euler's method; that is that we can show

$$|\mathcal{E}_N| \leq Kh^p \quad \text{for some} \quad p > 1.$$

Suppose we numerically solve some differential equation and estimate the error. If we think this error is too large we could redo the calculation with a smaller value of h. Or we could use a better method, for example **Runge-Kutta** methods. These are high-order methods that rely on evaluating $f(t, y)$ a number of times at each step in order to improve accuracy.

We'll first motivate one such method and then later look at the general framework.

The goal will be to develop some techniques to help us derive our own methods for accurately solving IVPs. Rather than using formal theory, we will reason based on carefully chosen examples.

5. Exercises

Exercise 2.3.1

An important step in the proof of Theorem 2.3.3, but which we didn't do in class, requires the observation that if $|\mathcal{E}_{i+1}| \leq |\mathcal{E}_i|(1+hL) + h|T_i|$, then

$$|\mathcal{E}_i| \leq \frac{T}{L}\left[(1+hL)^i - 1\right] \qquad i = 0, 1, \ldots, N.$$

Use induction to show that is indeed the case.

Exercise 2.3.2

Suppose we use Euler's method to find an approximation for $y(2)$, where y solves

$$y(1) = 1, \qquad y' = (t-1)\sin(y).$$

(i) Give an upper bound for the global error taking $N = 4$ (i.e., $h = 1/4$).

(ii) What N should you take to ensure that the global error is no more that 10^{-3}?